QVLA: Quantum-Inspired Wavefunction Kernel Attention with Tensor-Network Compression for Vision-Language-Action Models

VW Group Global Quantum + AI Challenge 2026 • Quantum-Enhanced Vision-Language-Action Models in Autonomous Driving and Robotics Applications

Problem Framing & Executive Proposition

Problem: Large VLAMs are difficult to train and deploy on embedded vehicle and robot hardware; quadratic attention also creates a real-time bottleneck. QVLA addresses two complementary constraints: attention efficiency and model footprint.
Proposition: QVLA combines WIKA / Q-WIKA for wavefunction-inspired efficient attention, with Q-WIKA providing a genuine QPU feature-map extension; MPS/TTN parameterization for selected Action-Decoder weights; and Flow Matching for continuous actions.

Challenge Alignment

Bottleneck	Response	Evidence
Attention Pairwise interaction grows approximately quadratically.	Multidimensional amplitude–phase WIKA state → overlap-inspired kernel → compact rank-r feature map; Q-WIKA tests compact quantum features.	Attention FLOPs, memory, wall-clock latency and task quality vs. strong classical controls.
Model footprint Large VLAMs challenge embedded deployment.	MPS/TTN factorization of selected Action-Decoder matrices; sweep bond dimension χ and fine-tune.	Parameters, size, memory, FLOPs, latency, compression ratio and quality.
Continuous action Realistic VLAM output pathway.	Flow-Matching Action Decoder with WIKA/Q-WIKA self-attention, cross-attention to H_{vl}, and MPS/TTN FFN.	Action error/success, trajectory quality, latency and reproducibility.

Scientific positioning. WIKA, MPS/TTN and other quantum-inspired components are classical implementations. No quantum advantage is assumed. Q-WIKA is the genuine quantum extension and will be evaluated with its execution and communication overhead.

Scope & Success Targets

Core scope excludes RL training, formal safety certification, interference-based fusion and measurement-inspired action gating. A compact VQC-ViT adapter is a secondary ablation only. Proposed goals are ≥10% efficiency improvement where applicable, ≤100 ms inference on the selected deployment profile where applicable, and ≥2× compression with ≤5% quality degradation for the compression pathway. At least three independent runs will be reported as mean ± standard deviation.

Technical Approach

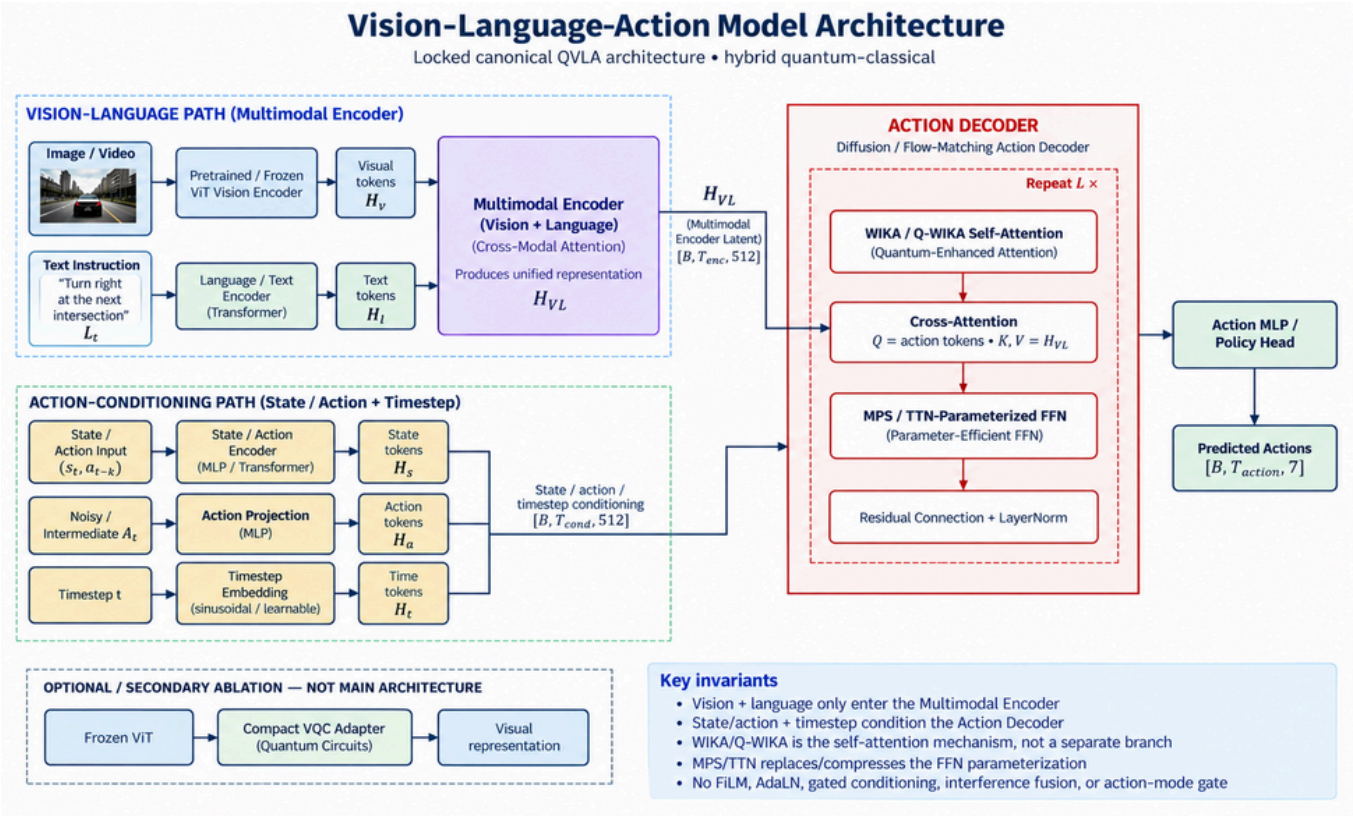


Figure 1. Canonical QVLA architecture. The action-conditioning connections terminate at the Action Decoder boundary; internal decoder blocks consume these signals.

Architecture & Data Flow

1. Input Encoding — VL Path

- Vision: Raw frames are converted into visual tokens via a frozen ViT.
- Language: Instructions are parsed into text tokens via a language encoder.
- Fusion: Visual and text tokens are combined in a Multimodal Encoder to produce H_{vl}.
- Dynamics: timestep t and action projection of A_t generate Action Tokens.

2. Decoder Block — Repeated L Times

- Utilizes WIKA / Q-WIKA mechanisms.
- Sequence: Self-Attention → Cross-Attention → MPS/TTN FFN → Residual Connections & Layer Normalization.
- Cross-Attention Routing: Action Tokens serve as the Queries (Q), while H_{vl} acts strictly as the external context through Keys/Values (K/V).
- H_{vl} is not directly mixed into the action-token sequence.

3. Output Head

- Decoded features pass through an Action MLP / Policy Head.
- Final control tensor: [B, T_{action}, 7]

WIKA / Q-WIKA

WIKA: Token/context features map to multidimensional complex states $\psi(x,c)=A(x,c)e^{i\varphi(x,c)}$. Amplitude captures context-dependent representational strength; phase is a learnable carrier for temporal, relational or cross-modal structure. Similarity uses $K(x,y)=|\langle\psi(x)|\psi(y)\rangle|^2$. Compact $\Phi(\psi)$ with rank $r \ll N$ avoids explicitly forming the full N×N interaction matrix.

QI-ARS: Rank selection is evaluated as an internal WIKA mechanism: fixed-r, spectral/effective-rank and task-aware r* controls are compared. The task-aware rule seeks the smallest r satisfying information-retention and task-performance thresholds.

Q-WIKA: A compact classical WIKA representation is encoded by $U(z)|0\ldots0\rangle$ on a QPU; measured observables/quantum-kernel features return compact quantum features. The QPU does not evaluate every token pair. Simulation → noise-aware simulation → real QPU is the planned pathway.

Methodology & Experimental Design

Flow Matching & Tensor-Network Compression

Flow Matching: Given a ground-truth action trajectory A, we construct an intermediate, noisy trajectory At A_t at a sampled flow time tt. We then project At A_t, combine it with timestep conditioning, and process it through the Action Decoder. The decoder predicts the target vector field at each action position; inference integrates this learned field to produce continuous actions. **MPS/TTN:** Selected dense Action-Decoder matrices are approximated as $W \approx \tilde{W}$. This approximation controls the compression–capacity trade-off. Candidate targets include FFN projections and selected attention projections. Fine-tuning is used to recover task quality where appropriate. The parameters rr and χ are distinct: r controls compact attention features, whereas χ controls tensorized weight representation.

Hybrid Quantum-Classical Workflow

Classical: data, frozen encoders, multimodal encoder, action conditioning, WIKA, compact feature processing, Action Decoder and Flow Matching. Quantum: optional Q-WIKA feature map and measurements on a QPU. The architecture is provider-neutral and does not require the entire VLAM to run on quantum hardware.

Optional Visual Ablation

A compact VQC adapter may be inserted after the pretrained/frozen ViT to study parameter-efficient visual adaptation. It is not part of the main architecture. Parameter reduction and total deployed-model compression will be reported separately.

Falsifiable Research Hypotheses

H1: Amplitude + phase WIKA improves useful multimodal similarity versus real-valued and amplitude-only controls. **H2:** Compact WIKA improves the accuracy–efficiency trade-off without full pairwise attention. **H3:** QI-ARS preserves task quality at lower average rank than fixed-r choices. **H4:** Q-WIKA features provide measurable value relative to their quantum overhead. **H5:** MPS/TTN reduces model footprint with acceptable task-quality loss.

Claims We Will Not Assume

The PoC will not assume that phase is inherently more informative, that quantum-inspired attention is always faster, that MPS/TTN automatically reduces latency, or that QPU execution provides quantum advantage. These are empirical questions tested against controlled baselines.

Validation Plan & Feasibility

ID	Validation / Ablation	Comparison
V1	Attention Baseline	Standard-attention VLAM (B0) vs. credible classical efficient-attention baseline
V2	WIKA Representation	Real-valued vs. amplitude-only vs. amplitude + phase
V3	Kernel Formulation	Dot-product attention vs. full WIKA vs. compact WIKA
V4	Adaptive Rank	Fixed-r vs. spectral/effective-rank vs. QI-ARS
V5	Quantum Extension	WIKA vs. Q-WIKA using simulator, noise-aware simulation, and QPU where available
V6	Tensor-Network Compression	Dense FFN vs. MPS/TTN across bond dimension χ
V7	Integrated QVLA	Best validated WIKA/QI-ARS + MPS/TTN + Q-WIKA components vs. B0
V8	Reproducibility	≥3 independent runs for each key configuration

Fair Comparison

Use standard attention plus a credible classical efficient-attention baseline. Match parameter count or total real degrees of freedom where practical. Measure wall-clock latency on identical hardware/software, not FLOPs alone. Isolate representation, kernel, rank, Q-WIKA and MPS/TTN effects. Attribute any quantum gain only when the quantum mechanism is isolated.

Metrics & Success Criteria

Task: success rate, action error, trajectory quality and applicable collision/near-miss or constraint metrics.
Efficiency: attention FLOPs, peak memory, throughput and end-to-end latency.
Compression: parameter count, serialized size, memory and compression ratio.
Quantum: qubit count, circuit depth, shots, execution/communication latency and noise sensitivity .**Targets**: ≥10% efficiency improvement and ≤100 ms inference on the selected deployment profile where applicable; ≥2× compression with ≤5% quality degradation for the compression pathway. Targets are not claimed results.

Resources & Feasibility

Use a reproducible autonomous-driving or robotics VLA benchmark with continuous actions; GPU compute for baseline and ablations; simulator first for Q-WIKA, followed by noise-aware simulation and real-QPU execution if accessible. PyTorch is the primary implementation environment; tensor-network tooling may include ITensor, quimb or tntorch.

Period	Focus	Expected Output
Weeks 1–3	Baseline, efficient-attention controls, reproducible evaluation harness and profiling	Baseline and evaluation harness
Weeks 4–7	WIKA representation, kernel and rank-r experiments	WIKA results and efficiency–quality curves
Weeks 8–10	MPS/TTN compression sweep, fine-tuning and integration	Compression operating points
Weeks 11–13	Q-WIKA simulation/noise-aware study; QPU test where available; end-to-end evaluation	Quantum overhead/benefit evidence
Weeks 14–16	Repeated runs, final profiling, reproducibility package and documentation	Final benchmark, code and report

Risks & Mitigations

WIKA gain absent: report negative result and retain classical comparator. Complex-state overhead: control real degrees of freedom and profile end-to-end. QPU overhead dominates: report quantum execution/communication costs separately. MPS/TTN quality loss: sweep χ and fine-tune. Deployment constraints: declare hardware/software configuration and use provider-neutral fallbacks.

Expected Enterprise Impact

The PoC will determine whether quantum-inspired attention and tensor-network parameterization can expand the practical operating envelope of VLAMs without requiring the entire model to run on quantum hardware. The direct objectives are lower attention cost, smaller model footprint, and quantified evidence for or against a QPU feature pathway.

Differentiation. WIKA addresses attention efficiency; MPS/TTN addresses model footprint; Q-WIKA creates a genuine quantum experiment; Flow Matching supplies a continuous-action interface. Each component is independently ablatable.

Deliverables

Reproducible QVLA implementation; standard and efficient-attention baselines; WIKA/QI-ARS ablations; MPS/TTN compression study; Q-WIKA simulation/noise-aware and, if feasible, QPU results; end-to-end latency/resource report; ≥3-run mean ± standard deviation results and documented configuration.

Team Profile & Execution Capability

Quantum Embodied Intelligence (QEI) • Team Lead: Mithilesh Nakade • 11-member multidisciplinary team. The profiles below summarize role, affiliation/background and directly relevant experience, as required by the Phase 1 guidelines.

Contect Email-ID: mithileshnakade@gmail.com (Team Lead)

Member	Role / Affiliation	Relevant Capability / Experience
Mithilesh Nakade	Lead • IIT Guwahati / University of Mumbai	AI/ML, DL, Transformers, data analytics; RL/OpenEnv; QVLA architecture.
Arumugam Selvakumar	Quantum • Anna Univ. / IIT Guwahati / IIT Delhi	15+ yrs software; quantum computing/QML; HHL, qubitization; quantum-AI.
Isparsh Chauhan	Physics-Informed AI • IIT Indore	CV, signal processing, predictive maintenance, physics-informed AI; physical systems.
Ujjwal Kumar	Algorithms & Optimization • Mahindra Univ. / IIT Guwahati	Algorithms, data structures, ML, optimization; efficiency and classical baselines.
Soumojay Karmakar	CV & Real-Time AI • MIT Bengaluru	CV, DL, NLP, Transformers, segmentation/detection; PyTorch/OpenCV; real-time video.
Purushotham Ganesh Kumar	Systems Integration • NMREC	~12.5 yrs software; APIs, microservices, enterprise architecture, performance.
Krish	Multimodal Data & Evaluation • Delhi SOL / IIT Guwahati	CV/NLP/RAG/ASR, data mining, sensor calibration; multimodal pipelines and evaluation.
Shivam Bhavsar	Generative & Agentic AI • IIT Guwahati / Masai	GenAI, LLMs, Transformers, RAG, agents; quantum-inspired wavefunction exploration.
Muneer Ahmad	ML & NLP • Islamia College / IIT Guwahati	AI/ML, DL, NLP, semantic embeddings, recommendation systems; project implementation.
Anurag Chakraborty	RL & AI • Sister Nivedita Univ. / IIT Guwahati	AI/ML, DL, reinforcement learning; enterprise software and SAP ABAP.
Deeksha Chaturvedi	AI Applications & Edge • Poornima / Univ. Rajasthan / IIT Guwahati	~9.5 yrs mobile development; NLP/RAG; Android, iOS, React Native; edge integration.

Collective Execution Capability

Quantum & hybrid AI: dedicated quantum-computing/algorithm expertise with AI/Transformer architecture. VLAM & multimodal: vision, language, RAG, semantic representation and evaluation. Efficiency: algorithms, optimization, classical controls and tensor-network experimentation. Physical AI: physics-informed AI, signal processing and engineering-domain perspective. Systems & deployment: enterprise architecture, performance engineering, real-time vision and mobile/edge development. Execution fit. QEI combines the complementary skills needed to implement the canonical QVLA architecture, establish strong classical controls, conduct quantum-inspired and QPU experiments, compress the Action Decoder, and evaluate an end-to-end continuous-action VLAM. Individual claims are intentionally conservative; the execution capability is based on the combined team.

Quantum+VLA Value Proposition

Challenge	QVLA Contribution	Potential Enterprise Value
High attention cost	WIKA / compact WIKA	Lower attention compute and memory demand
Large model footprint	MPS/TTN parameterization	Smaller deployable model footprint
Continuous control generation	Flow-Matching Action Decoder	Natural continuous action/trajectory generation
Quantum-AI uncertainty	Q-WIKA with controlled quantum experiments	Evidence-based assessment of quantum feature pathways
Embedded deployment constraints	Efficiency + compression + latency evaluation	Better understanding of practical deployment envelope

Final Outcome

A validated, benchmarked pathway for determining whether quantum-inspired attention, tensor-network compression, and quantum feature maps can improve the efficiency and deployment feasibility of continuous-action VLAMs.